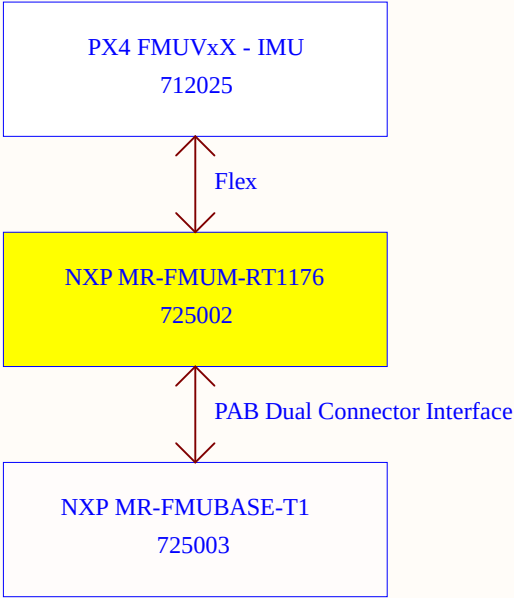


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8	POWER
9	MEMORY

REVISIONS				
Authority	Rev	Description	Date	Approved
ENGR	01	Initial Release	11/03/2022	DBS
ENGR	01	Delete C74, C90, C93 DCDC_DIG_1V0 0.1uF caps. Connect SE051_ENA thru N-FET to HW_VER_REV_DRIVE. Added grounds to J4-24, J4-34. Add 0 ohm DNP between SPI6 MOSI and SCK. Change R35 to 4.7K always populated. Change BOOT_MODE1 connection to GND from direct to 10K pull down. Remove TRACE PUs/PDs. Added TP to TRACESWO. Removed TP on USB2_VBUS. Changed table on page 6. Change Enable to SE050 to use HW_VER_REV_DRIVE and a FET. Change VDD_1V8 input to VDD_5V_IN. Populate all ADC scaling resistors on page 8. Tie FMU_nRST to 1V8 HyperFlash through 0 ohm resistor. Change from ICM-20602 to ICM-42688-P. Moved C22, C23, U3 away from connector J3. Moved J5 1.25mm from board edge. Change X1 to Hongyun part, U13 to SOIC-8, L1 to SWPA4018S4R7MT. Added 3x 0.1uF FMU_VDD_3V3 caps. Swap pinnings of SPI3_DRDY2_SENSOR3 w/ BUZZER_1, FMU_PPM_INPUT w/ GPIO_EMC_B2_12 (was GPIO_EMC_B1_09).	4/5/2023	DBS
ENGR	02	Add 2x 11mF caps B2, B3 as pop option to replace B1 70mF obsolete cap (sh 8). Delete diodes between supercap and V_RTC_3V3, moved to Base. Other cleanup.	6/11/2024	DBS



NOTES, unless otherwise specified

- All Resistors 1%, 1/16 W
- All Capacitors are in micro Farads, +/- 10%, 50V

NOTE: This schematic needs to be viewed in color.

Text in **BLUE** are not net names. They are labels to reference the DS-010 Pixhawk Autopilot Bus (PAB) Standard.

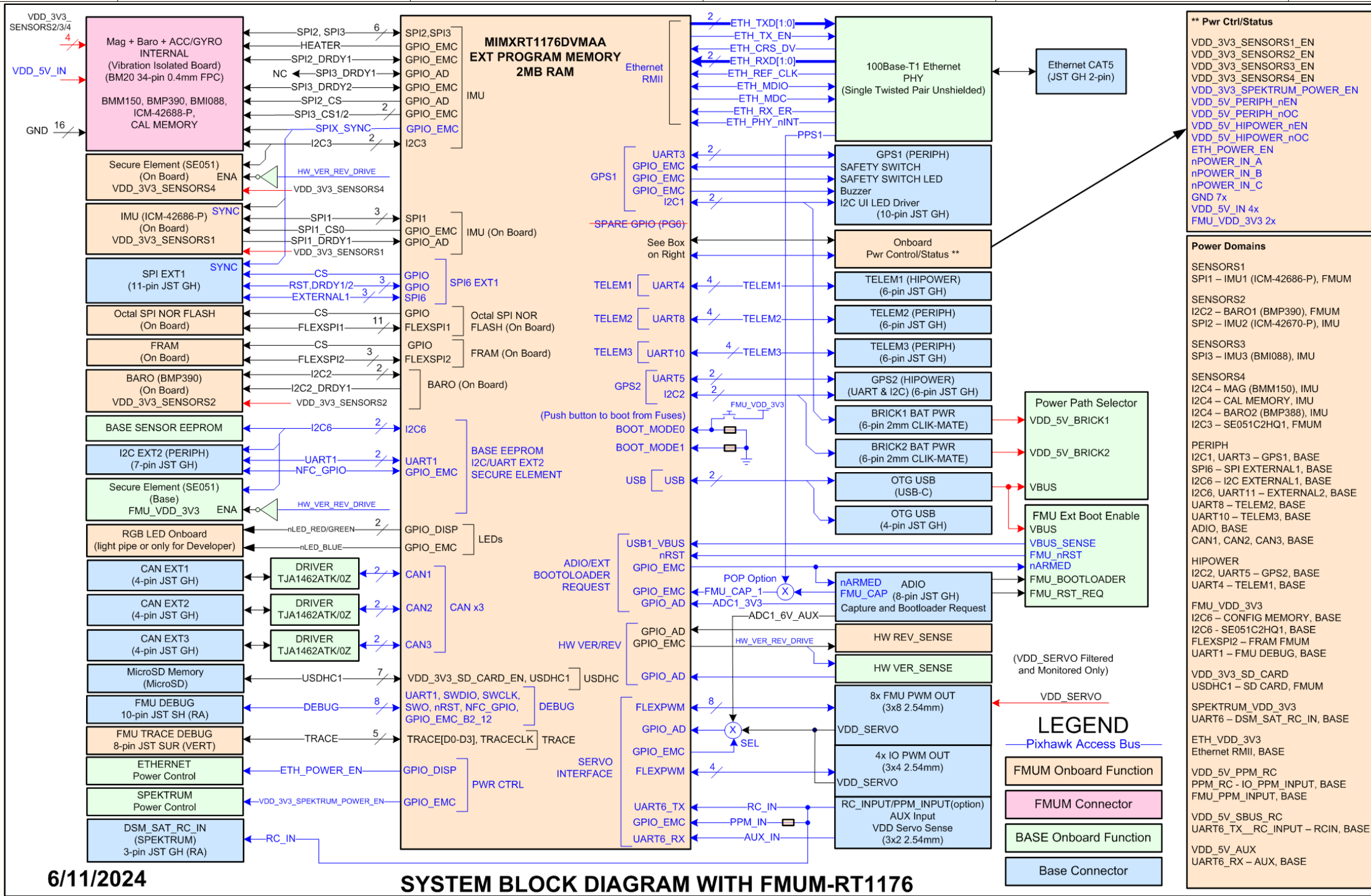
REF DES	ASSY OPT	PAGE NAME
R64	DNP	03_PAB
R13, R17, J1	DNP	04_FMU_MCU1
R49	DNP	07_SENSORS



System Innovation, Mobile Robotics Domain

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Designer: Dugan Yoon	Drawing Title: MR-FMUM-RT1176		
Checked: David Sidrane	Page Title: 01_cover.SchDoc		
Approved: Iain Galloway	Size B	Document Number 725002	Rev 02
Date: 6/11/2024			Sheet 1 of 9



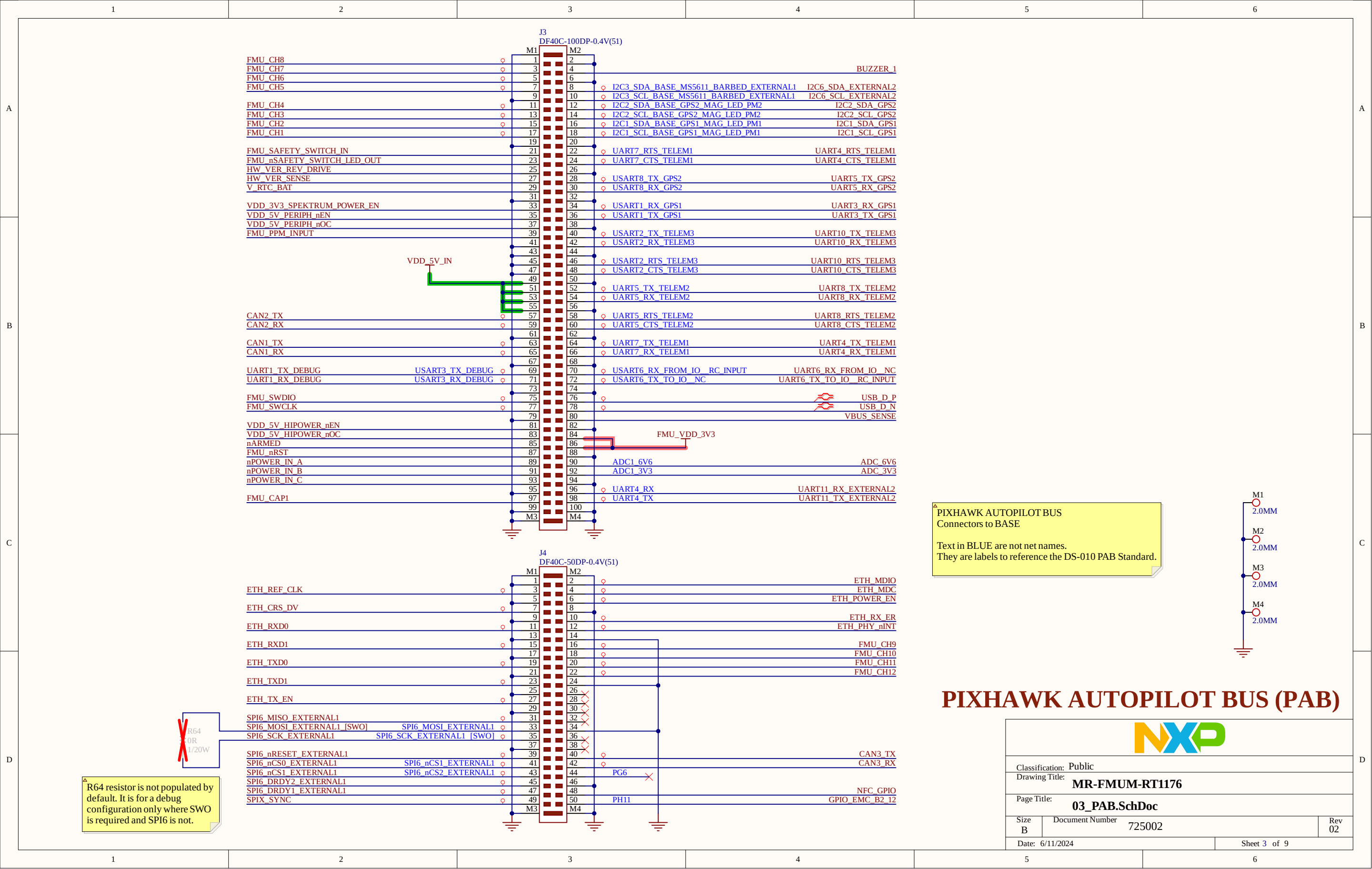
6/11/2024

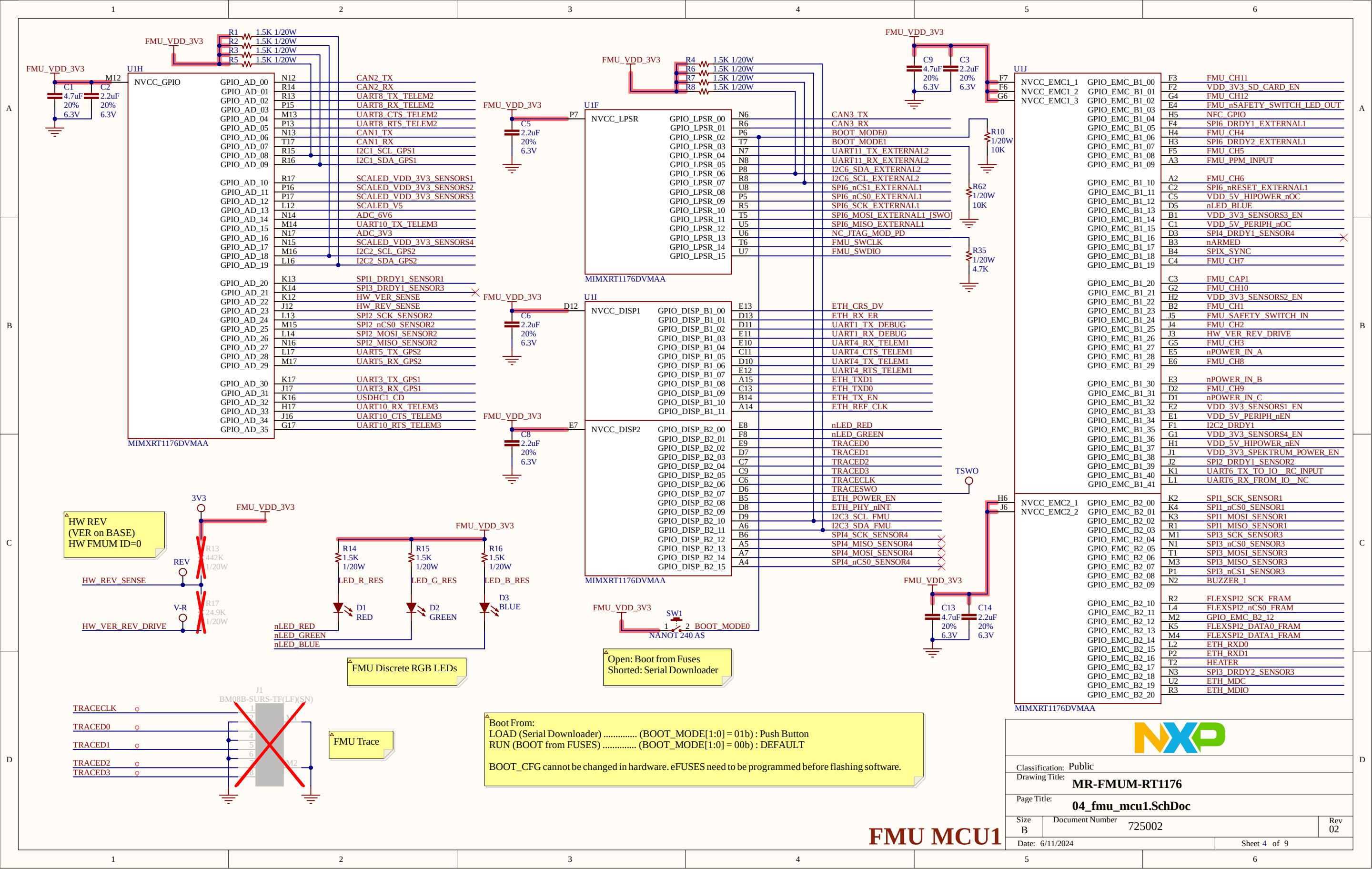
SYSTEM BLOCK DIAGRAM WITH FMUM-RT1176

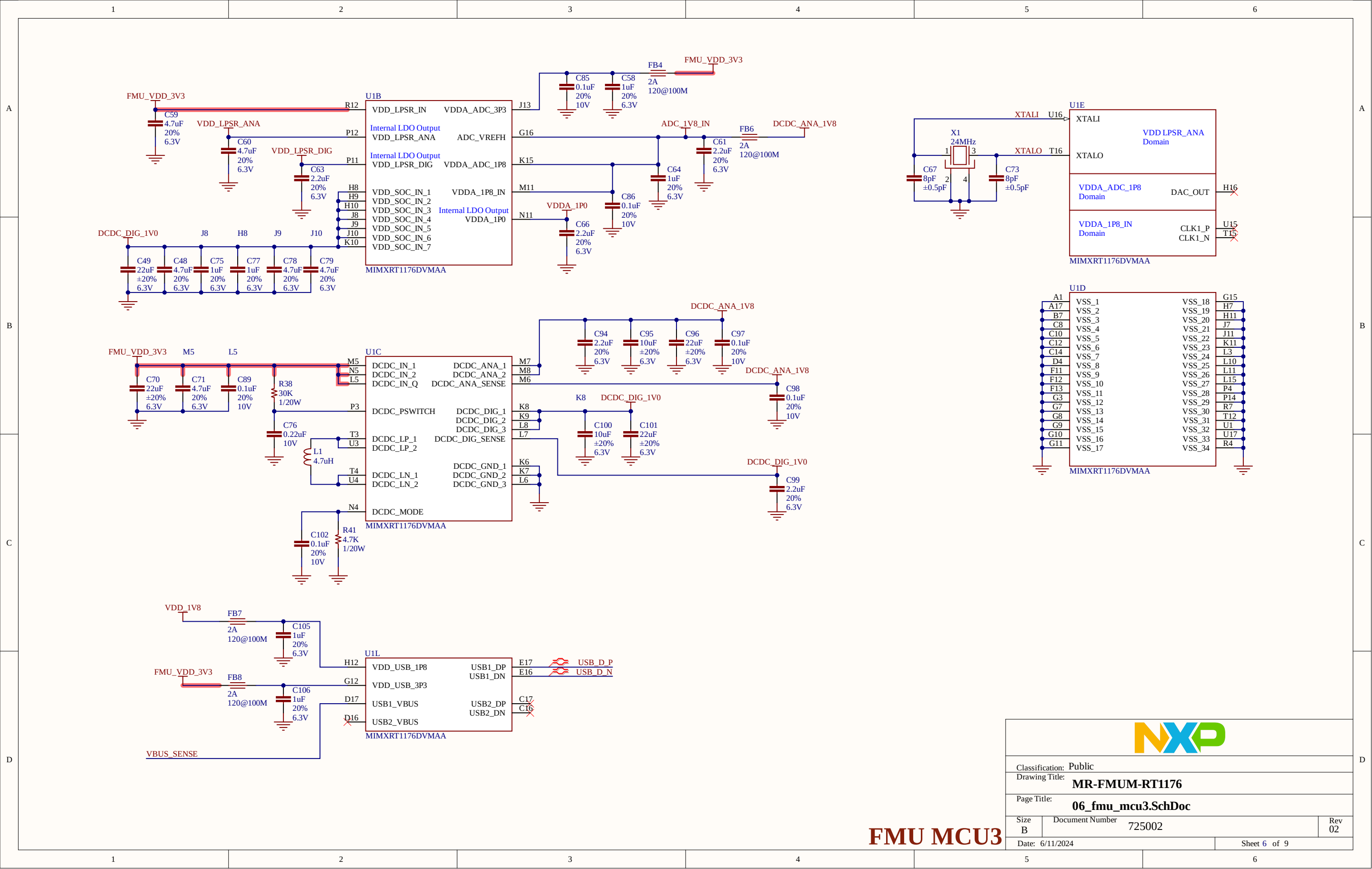


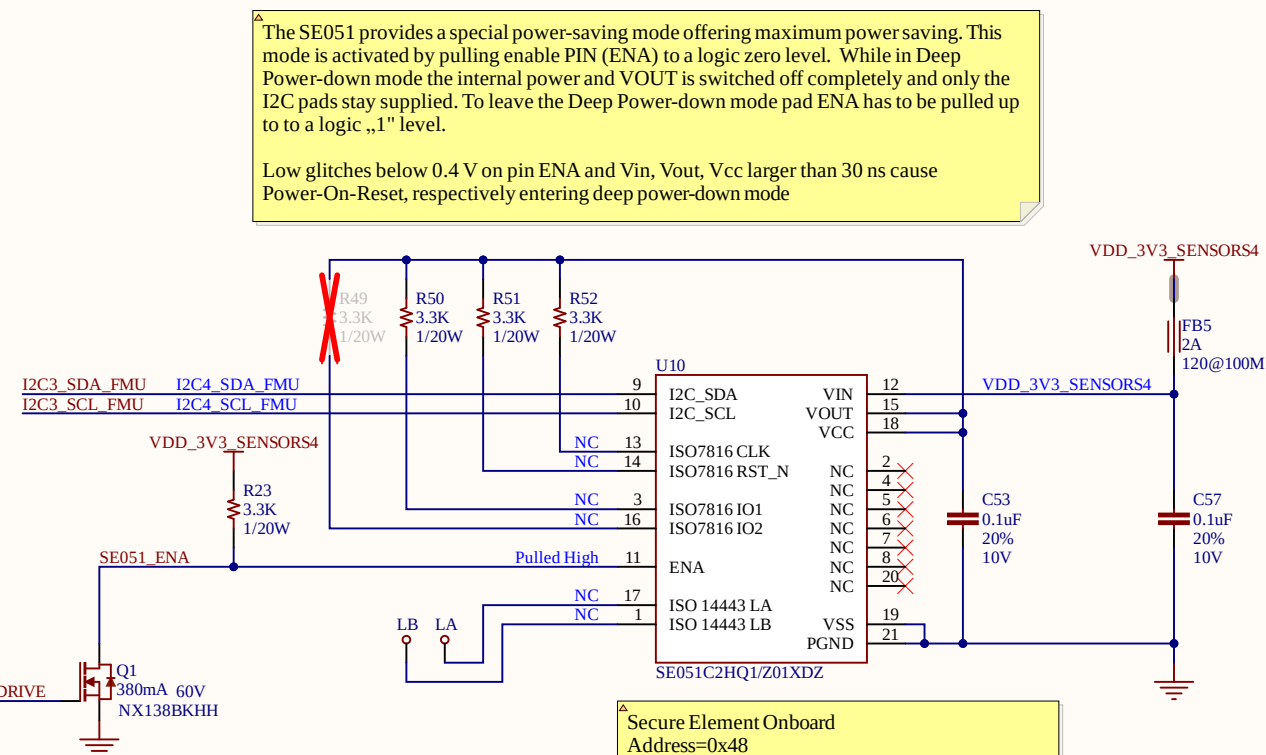
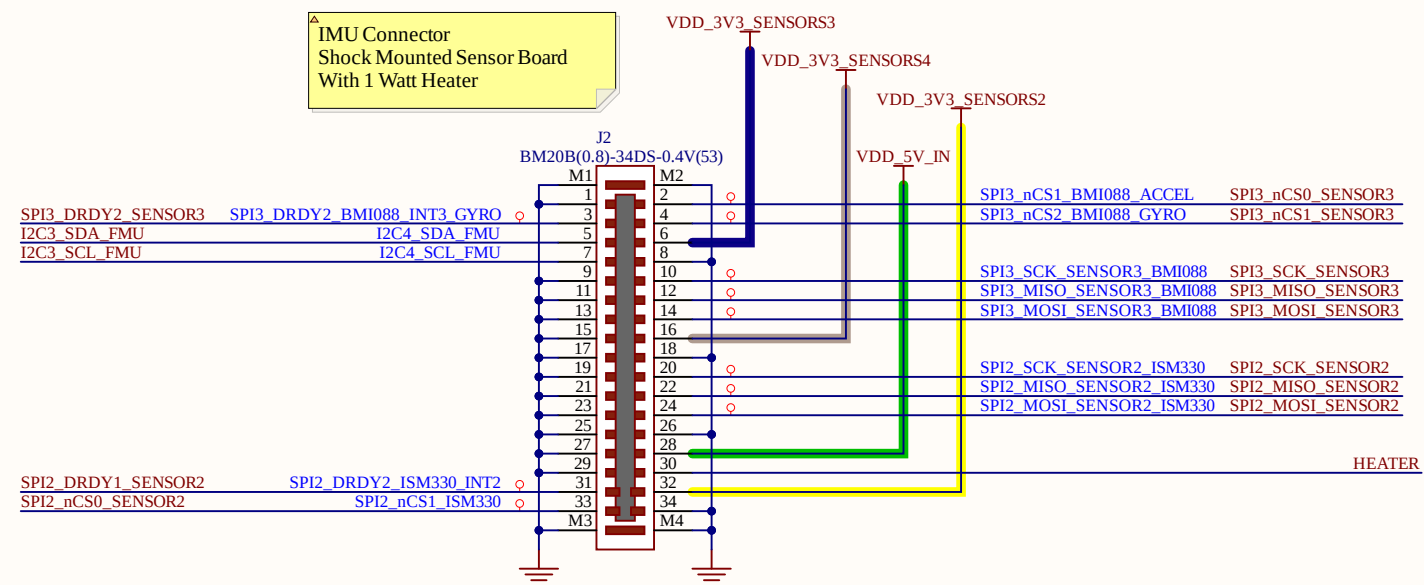
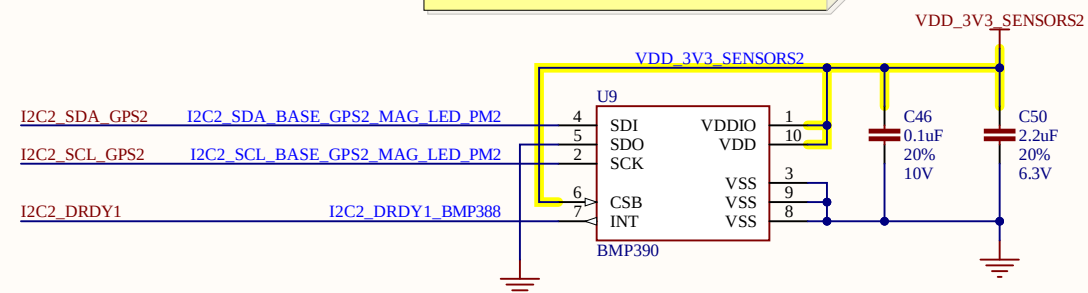
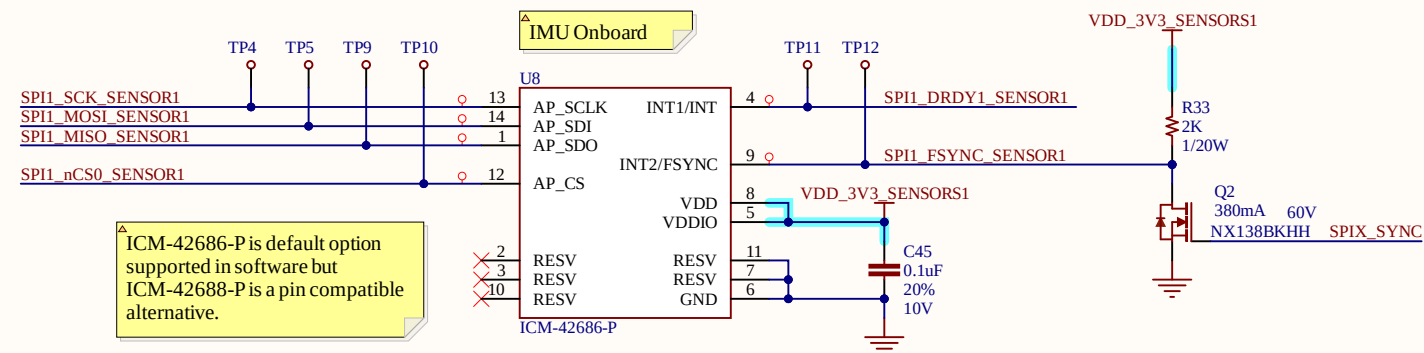
Classification: Public		
Drawing Title: MR-FMUM-RT1176		
Page Title: 02_block_diagram.SchDoc		
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BLOCK DIAGRAM









△ To enable SE050 drive HW_VER_REV_DRIVE low.

Power on sequence:

- 1) Drive I2C4 lines to output low (avoid backfeeding SE050)
- 2) Do HW versioning with VDD_3V3_SENSORS4 off. Leave HW_VER_REV_DRIVE high (SE050 disabled) on exit.
- 3) Then set HW_VER_REV_DRIVE low (SE050 enabled).
- 4) Then power on VDD_3V3_SENSORS4.
- 5) HW_VER_REV_DRIVE can be used to toggle SE051_ENA later if needed.

Secure Element Onboard
Address=0x48
Text in BLUE are not net names.
They are labels to reference the v6X pinout.

A

B

C

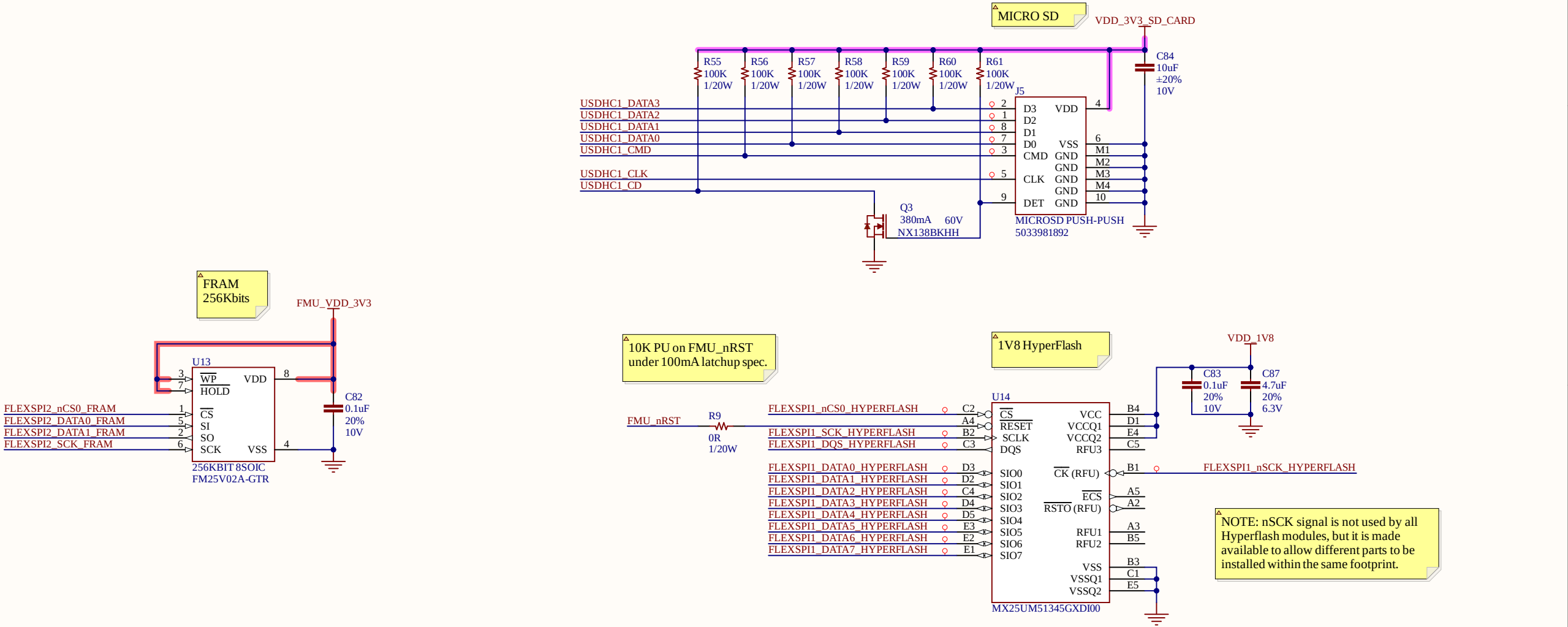
D

A

B

C

D



MEMORY



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